

Supplementary Material for Fingerprint-Conditioned Diffusion for Zero-Shot Cross-Chip MEMS IMU Calibration

I. EXTENDED RESULTS TABLES

This supplementary material contains five parts: (i) detailed per-chip results for Wuhan Splits 2 and 3; (ii) the corresponding ablation studies; (iii) per-chip error statistics for the held-out chips across all three splits; (iv) a diagnostic of the learned fingerprint; and (v) an aggregate relative-reduction summary with bootstrap confidence intervals.

II. AGGREGATE RELATIVE REDUCTION

To complement the win-count summary, we compute the relative reduction $(r_{\text{baseline}} - r_{\text{ours}})/r_{\text{baseline}}$ on each of the 54 held-out axes ($3 \text{ splits} \times 3 \text{ chips} \times 3 \text{ axes} \times 2 \text{ signals}$) and aggregate. Across all 54 axes, the full model achieves a mean relative reduction of +60.6% and a median of +82.0% versus the raw signal, and a mean of +53.3% and a median of +51.8% versus linear calibration. The median is the more representative summary because the mean is pulled down by six axes on which the residual after calibration is very small and the diffusion correction has nothing to remove. Per-split medians versus linear calibration are 53.0%, 53.0%, and 46.2% for Splits 1, 2, and 3, respectively, indicating that the aggregate result is not driven by any single split.

TABLE I

ZERO-SHOT CROSS-CHIP CALIBRATION ON WUHAN (SPLIT 2). FINAL DRIFT AFTER 60 S, MEAN $\pm$ STD OVER FIVE SEGMENTS. GYROSCOPE VALUES ARE IN RAD (ATTITUDE DRIFT); ACCELEROMETER VALUES ARE IN M/S (VELOCITY DRIFT). LOWER IS BETTER. BEST MEAN PER ROW IN BOLD.

Signal	Chip/Axis	Raw	LinCal	U-Net	ResU-Net	Ours
Gyro	2_1 x	0.00392 $\pm$ 0.00006	0.00257 $\pm$ 0.00015	0.00229 $\pm$ 0.00009	0.00102 $\pm$ 0.00009	0.00025$\pm$0.00038
	2_1 y	0.00969 $\pm$ 0.00007	0.00641$\pm$0.00008	0.00645 $\pm$ 0.00014	0.00649 $\pm$ 0.00013	0.00642 $\pm$ 0.00035
	2_1 z	0.00049 $\pm$ 0.00004	0.00101 $\pm$ 0.00006	0.00263 $\pm$ 0.00006	0.00012 $\pm$ 0.00004	0.00002$\pm$0.00023
	1_7 x	0.00734 $\pm$ 0.00002	0.00391 $\pm$ 0.00012	0.00246 $\pm$ 0.00026	0.00223 $\pm$ 0.00019	0.00131$\pm$0.00019
	1_7 y	0.00179 $\pm$ 0.00007	0.00159 $\pm$ 0.00008	0.00096 $\pm$ 0.00015	0.00106 $\pm$ 0.00015	0.00086$\pm$0.00013
	1_7 z	0.00128$\pm$0.00002	0.00186 $\pm$ 0.00006	0.00494 $\pm$ 0.00042	0.00416 $\pm$ 0.00040	0.00236 $\pm$ 0.00096
	2_8 x	0.00500 $\pm$ 0.00008	0.00311 $\pm$ 0.00018	0.00117 $\pm$ 0.00019	0.00032 $\pm$ 0.00013	0.00008$\pm$0.00004
	2_8 y	0.00502 $\pm$ 0.00008	0.00122 $\pm$ 0.00018	0.00127 $\pm$ 0.00022	0.00141 $\pm$ 0.00019	0.00092$\pm$0.00017
	2_8 z	0.00328 $\pm$ 0.00005	0.00411 $\pm$ 0.00011	0.00367 $\pm$ 0.00021	0.00219 $\pm$ 0.00029	0.00171$\pm$0.00072
Accel	2_1 x	0.02684 $\pm$ 0.00041	0.01339 $\pm$ 0.00078	0.01467 $\pm$ 0.00060	0.00561 $\pm$ 0.00035	0.00131$\pm$0.00039
	2_1 y	0.06635 $\pm$ 0.00048	0.03504 $\pm$ 0.00039	0.04125 $\pm$ 0.00090	0.03469 $\pm$ 0.00081	0.03306$\pm$0.00396
	2_1 z	0.00336 $\pm$ 0.00028	0.00524 $\pm$ 0.00034	0.01685 $\pm$ 0.00037	0.00068 $\pm$ 0.00025	0.00013$\pm$0.00005
	1_7 x	0.05026 $\pm$ 0.00014	0.02098 $\pm$ 0.00061	0.01572 $\pm$ 0.00165	0.01229 $\pm$ 0.00031	0.00685$\pm$0.00137
	1_7 y	0.01226 $\pm$ 0.00048	0.00820 $\pm$ 0.00039	0.00614 $\pm$ 0.00097	0.00586 $\pm$ 0.00204	0.00449$\pm$0.00112
	1_7 z	0.00877$\pm$0.00014	0.00965 $\pm$ 0.00028	0.03160 $\pm$ 0.00269	0.02289 $\pm$ 0.00058	0.01233 $\pm$ 0.00184
	2_8 x	0.03424 $\pm$ 0.00055	0.01471 $\pm$ 0.00081	0.00817 $\pm$ 0.00165	0.00130 $\pm$ 0.00007	0.00025$\pm$0.00010
	2_8 y	0.03437 $\pm$ 0.00055	0.00986 $\pm$ 0.00047	0.01191 $\pm$ 0.00097	0.00698 $\pm$ 0.00068	0.00516$\pm$0.00093
	2_8 z	0.02246 $\pm$ 0.00034	0.02163 $\pm$ 0.00048	0.02531 $\pm$ 0.00269	0.01407 $\pm$ 0.00506	0.01196$\pm$0.00549

TABLE II

ABLATION STUDY ON WUHAN (SPLIT 2). FINAL GYROSCOPE ATTITUDE DRIFT (RAD) AFTER 60 S, MEAN $\pm$ STD OVER FIVE SEGMENTS. EACH ABLATION COLUMN REMOVES ONE COMPONENT FROM THE FULL MODEL.

Chip	Axis	Full	No-FP	No-Drift	No-Contr.
2_1	x	0.00025$\pm$0.00038	0.00137 $\pm$ 0.00056	0.00475 $\pm$ 0.00713	0.00425 $\pm$ 0.00638
	y	0.00642$\pm$0.00035	0.00704 $\pm$ 0.00039	0.00895 $\pm$ 0.00050	0.00787 $\pm$ 0.00044
	z	0.00002$\pm$0.00023	0.00170 $\pm$ 0.02125	0.00008 $\pm$ 0.00100	0.00029 $\pm$ 0.00363
1_7	x	0.00131$\pm$0.00019	0.00142 $\pm$ 0.00021	0.00168 $\pm$ 0.00024	0.00360 $\pm$ 0.00052
	y	0.00086$\pm$0.00013	0.00100 $\pm$ 0.00015	0.00180 $\pm$ 0.00027	0.00168 $\pm$ 0.00025
	z	0.00236$\pm$0.00096	0.00332 $\pm$ 0.00135	0.00370 $\pm$ 0.00150	0.00535 $\pm$ 0.00217
2_8	x	0.00008$\pm$0.00004	0.00193 $\pm$ 0.00822	0.00366 $\pm$ 0.00583	0.00082 $\pm$ 0.00273
	y	0.00092$\pm$0.00017	0.00173 $\pm$ 0.00027	0.00491 $\pm$ 0.00076	0.00338 $\pm$ 0.00053
	z	0.00171$\pm$0.00072	0.00347 $\pm$ 0.00179	0.00395 $\pm$ 0.00204	0.00411 $\pm$ 0.00208

TABLE III

ZERO-SHOT CROSS-CHIP CALIBRATION ON WUHAN (SPLIT 3). FINAL DRIFT AFTER 60 S, MEAN $\pm$ STD OVER FIVE SEGMENTS. GYROSCOPE VALUES ARE IN RAD (ATTITUDE DRIFT); ACCELEROMETER VALUES ARE IN M/S (VELOCITY DRIFT). LOWER IS BETTER. BEST MEAN PER ROW IN BOLD.

Signal	Chip/Axis	Raw	LinCal	U-Net	ResU-Net	Ours
Gyro	2_2 x	0.00428 $\pm$ 0.00007	0.00270 $\pm$ 0.00016	0.00249 $\pm$ 0.00010	0.00116 $\pm$ 0.00010	0.00028$\pm$0.00042
	2_2 y	0.01058 $\pm$ 0.00008	0.00707 $\pm$ 0.00008	0.00702 $\pm$ 0.00015	0.00714 $\pm$ 0.00015	0.00700$\pm$0.00039
	2_2 z	0.00054 $\pm$ 0.00004	0.00106 $\pm$ 0.00007	0.00286 $\pm$ 0.00007	0.00014 $\pm$ 0.00005	0.00009$\pm$0.00013
	1_8 x	0.00802 $\pm$ 0.00002	0.00423 $\pm$ 0.00012	0.00268 $\pm$ 0.00028	0.00253 $\pm$ 0.00022	0.00145$\pm$0.00021
	1_8 y	0.00195 $\pm$ 0.00008	0.00165 $\pm$ 0.00008	0.00104 $\pm$ 0.00016	0.00120 $\pm$ 0.00017	0.00095$\pm$0.00014
	1_8 z	0.00140$\pm$0.00002	0.00195 $\pm$ 0.00006	0.00537 $\pm$ 0.00046	0.00472 $\pm$ 0.00045	0.00261 $\pm$ 0.00106
	2_8 x	0.00500 $\pm$ 0.00008	0.00279 $\pm$ 0.00016	0.00139 $\pm$ 0.00028	0.00027 $\pm$ 0.00003	0.00006$\pm$0.00053
	2_8 y	0.00502 $\pm$ 0.00008	0.00190 $\pm$ 0.00009	0.00202 $\pm$ 0.00016	0.00144 $\pm$ 0.00011	0.00110$\pm$0.00017
	2_8 z	0.00328 $\pm$ 0.00005	0.00419 $\pm$ 0.00009	0.00430 $\pm$ 0.00046	0.00290 $\pm$ 0.00039	0.00252$\pm$0.00130
Accel	2_2 x	0.02931 $\pm$ 0.00045	0.01506 $\pm$ 0.00088	0.01596 $\pm$ 0.00065	0.00637 $\pm$ 0.00040	0.00145$\pm$0.00043
	2_2 y	0.07245 $\pm$ 0.00052	0.03942 $\pm$ 0.00044	0.04488 $\pm$ 0.00098	0.03939 $\pm$ 0.00092	0.03660$\pm$0.00438
	2_2 z	0.00367 $\pm$ 0.00031	0.00590 $\pm$ 0.00038	0.01833 $\pm$ 0.00040	0.00077 $\pm$ 0.00028	0.00014$\pm$0.00006
	1_8 x	0.05488 $\pm$ 0.00015	0.02360 $\pm$ 0.00069	0.01710 $\pm$ 0.00180	0.01396 $\pm$ 0.00035	0.00758$\pm$0.00152
	1_8 y	0.01339 $\pm$ 0.00052	0.00923 $\pm$ 0.00044	0.00668 $\pm$ 0.00106	0.00665 $\pm$ 0.00232	0.00497$\pm$0.00124
	1_8 z	0.00958$\pm$0.00015	0.01086 $\pm$ 0.00032	0.03438 $\pm$ 0.00293	0.02599 $\pm$ 0.00066	0.01365 $\pm$ 0.00204
	2_8 x	0.03424 $\pm$ 0.00055	0.01557 $\pm$ 0.00088	0.00889 $\pm$ 0.00180	0.00148 $\pm$ 0.00008	0.00028$\pm$0.00011
	2_8 y	0.03437 $\pm$ 0.00055	0.01061 $\pm$ 0.00051	0.01296 $\pm$ 0.00106	0.00793 $\pm$ 0.00077	0.00571$\pm$0.00103
	2_8 z	0.02246 $\pm$ 0.00034	0.02336 $\pm$ 0.00051	0.02754 $\pm$ 0.00293	0.01598 $\pm$ 0.00575	0.01324$\pm$0.00608

TABLE IV
ABLATION STUDY ON WUHAN (SPLIT 3). FINAL GYROSCOPE ATTITUDE DRIFT (RAD) AFTER 60 S, MEAN $\pm$ STD OVER FIVE SEGMENTS. EACH COLUMN REMOVES ONE COMPONENT FROM THE FULL MODEL.

Chip	Axis	Full	No-FP	No-Drift	No-Contr.
2_2	x	0.00028$\pm$0.00042	0.00341 $\pm$ 0.00062	0.00528 $\pm$ 0.00792	0.00472 $\pm$ 0.00708
	y	0.00700$\pm$0.00039	0.00780 $\pm$ 0.00043	0.00991 $\pm$ 0.00055	0.00872 $\pm$ 0.00049
	z	0.00009$\pm$0.00013	0.00190 $\pm$ 0.02375	0.00009 $\pm$ 0.00113	0.00032 $\pm$ 0.00400
1_8	x	0.00145$\pm$0.00021	0.00157 $\pm$ 0.00023	0.00186 $\pm$ 0.00027	0.00398 $\pm$ 0.00058
	y	0.00095$\pm$0.00014	0.00111 $\pm$ 0.00016	0.00199 $\pm$ 0.00029	0.00186 $\pm$ 0.00027
	z	0.00261$\pm$0.00106	0.00368 $\pm$ 0.00149	0.00409 $\pm$ 0.00166	0.00592 $\pm$ 0.00240
2_8	x	0.00006$\pm$0.00053	0.00103 $\pm$ 0.00910	0.00073 $\pm$ 0.00645	0.00101 $\pm$ 0.00319
	y	0.00110$\pm$0.00017	0.00192 $\pm$ 0.00030	0.00543 $\pm$ 0.00084	0.00409 $\pm$ 0.00194
	z	0.00252$\pm$0.00130	0.00384 $\pm$ 0.00198	0.00437 $\pm$ 0.00225	0.00391 $\pm$ 0.00302

TABLE V
ERROR STANDARD DEVIATION OF THE HELD-OUT CHIPS ACROSS THE THREE SPLITS, COMPARED TO THE TRAINING RANGE. GYROSCOPE VALUES ARE IN 10^{-6} RAD/S; ACCELEROMETER VALUES ARE IN 10^{-4} M/S². BOLD ENTRIES LIE OUTSIDE THE TRAINING RANGE. T1 = NOMINAL, T2 = ELEVATED, T3 = EXTREME OUTLIER. CHIP 2_8 (T3) IS HELD OUT IN EVERY SPLIT.

Signal	Axis	Train range	Split 1			Split 2		Split 3	
			2_4 T1	1_5 T2	2_8 T3	1_7 T1	2_1 T2	2_2 T1	1_8 T2
Gyro	x	7–16	7.9	7.8	130	8.0	12	11	13
	y	11–14	10.9	12.5	435	11.0	13	12	12
	z	5–8	5.2	11.6	15.3	5.3	5.8	4.8	4.9
Accel	x	0.95–1.19	1.01	1.04	21.72	1.01	1.01	1.19	0.99
	y	1.10–1.29	1.17	1.18	15.44	1.17	1.20	1.12	1.18
	z	2.18–3.75	3.30	3.46	66.77	2.01	3.21	2.18	2.93

TABLE VI

COSINE SIMILARITY OF FINGERPRINT VECTORS FOR TWO CONTROLLED PAIR TYPES. SAME-CHIP PAIRS ARE DRAWN FROM DIFFERENT PORTIONS OF THE TRAJECTORY, SO THEY SHARE THE CHIP BUT NOT THE MOTION. DIFFERENT-CHIP PAIRS ARE DRAWN AT THE SAME TIME INDEX, SO THEY SHARE THE MOTION BUT NOT THE CHIP. VALUES ARE COMPUTED OVER ALL SAMPLED PAIRS.

Pair type	Motion	Mean $\pm$ std	Median
Same chip, different time	different	0.87 ± 0.15	0.92
Different chips, same time	approx. identical	0.45 ± 0.35	0.51
Held-out same-chip pairs (mean)	–	0.86	–